

SCHOOL OF INFORMATION AND COMMUNICATION TECHNOLOGY

DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING

COURSE STRUCTURE

Under New Education Policy (NEP) 2020: Academic Year 2026-2027 and Onwards
B.Tech. (Computer Science and Engineering) and Specializations: Data Science, Cyber Security
Integrated B.Tech.- M.Tech. (Computer Science and Engineering)



After Year 4: B.Tech. Degree

After Year 3: B.Sc. (Computer Science)

After Year 2: UG Diploma

After Year 1: UG
Certificate

SEMESTER V										UCC Mapping	
S.No	Course Code	Course Name	L	T	P	Credits	UCC Mapping	UCC Mapping	UCC Mapping	UCC Mapping	
1	CS401	Computer Design	3	0	0	3	PEC	Professional Core Course	Major (DSG)	Major (DSG)	
2	CS402	Software Engineering	3	0	0	3	PEC	Professional Core Course	Major (DSG)	Major (DSG)	
3	CS403	High Performance Computing (HPC)	3	0	0	3	PEC	Professional Core Course	Major (DSG)	Major (DSG)	
4	CS404	Specialization Core Course - II	3	0	0	3	PEC	Professional Core Course	Major (DSG)	Major (DSG)	
5	CS405	Specialization Core Course - III	3	0	0	3	PEC	Professional Core Course	Major (DSG)	Major (DSG)	
6	CS406	Program Elective 1	3	0	0	3	PEC	Professional Core Course	Major (DSG)	Major (DSG)	
7	CS407	Program Elective 2	3	0	0	3	PEC	Professional Core Course	Major (DSG)	Major (DSG)	
8	CS408	Specialization Core Course - III Lab	0	0	4	2	PEC	Professional Core Course	Major (DSG)	Major (DSG)	
9	CS409	Program Elective 1 Lab	0	0	4	2	PEC	Professional Core Course	Major (DSG)	Major (DSG)	
Total Hours and Credits			18	0	8	22					

SEMESTER VI							ACTING MAPING	
	Course Code	Course Name	L	T	P	Credits		Major (DSG)
1	CS501	Internet of Things	3	0	0	3	PEC	Engineering Science Course
2	CS502	Cloud Infrastructure and Virtualization	3	0	0	3	PEC	Engineering Science Course
3	CS503	Unmanned Aerial Vehicle System Engineering	3	0	0	3	PEC	Engineering Science Course
4	CS504	Program Elective 3	3	0	0	3	PEC	Professional Elective Course
5	CS505	Program Elective 4	3	0	0	3	PEC	Professional Elective Course
6	CS506	Open Elective 1	3	0	0	3	PEC	Open Elective Course
7	CS507	Open Elective 2	3	0	0	3	PEC	Open Elective Course
8	CS508	Internet of Things Lab	0	0	4	2	PEC	Engineering Science Course
9	CS509	Cloud Infrastructure and Virtualization Lab	0	0	4	2	PEC	Engineering Science Course
10	CS510	Capstone Project 1*	0	0	4	2	Project	Research Project, Major
Total Hours and Credits			18	0	12	24		

* The Capstone Project-1/Capstone Project-2 will be completed in a group, and it will also be evaluated as per University Examination Rules.

SEMESTER VII							UCC Mapping		
Course Code		Course Name	L	T	P	Credits	UCC Mapping		
1	XXXXX	Research Methodology	3	0	0	3	PEC	Professional Core Course	Major (DSG)
2	XXXXX	Open Elective 2	3	0	0	3	PEC	Open Elective Course	Major (DSG)
3	XXXXX	Open Elective 3	3	0	0	3	PEC	Open Elective Course	Major (DSG)
4	XXXXX	Specialization Core Course - I/II/III	3	0	0	3	UAMC	Major	Major (DSG)
5	XXXXX	Capstone Project 2*	0	0	4	4	Project	Project	Major
Total Hours and Credits			12	0	8	16			

SEMESTER VIII										UCC Mapping	
S.No	Course Code	Course Name	L	T	P	Credits	UCC Mapping	UCC Mapping	UCC Mapping	UCC Mapping	
1	CS701	Internship 1	0	0	20	10	Internship	Internship	Major (DSG)	Major (DSG)	
2	CS702	Internship 2	0	0	20	10	Internship	Internship	Major (DSG)	Major (DSG)	
Total Hours and Credits			0	0	40	20					

* Internship will be done during the summer vacation period during the 5th semester in industry / Govt. These assignments and report submission will be for a minimum of 6 months. It will be evaluated as per the University Examination rules.

Total Credits:

168

SPECIALIZATION CORE COURSES - I/II/III (Computer Science and Engineering)										UCC Mapping	
S.No	Course Code	Course Name	L	T	P	Credits	UCC Mapping	UCC Mapping	UCC Mapping	UCC Mapping	UCC Mapping
1	CS801	Machine Learning	3	0	0	3	PEC	Specialization Core Course	Major (DSG)	Major (DSG)	Major (DSG)
2	CS802	Machine Learning Lab	0	0	4	2	PEC	Specialization Core Course	Major (DSG)	Major (DSG)	Major (DSG)
3	CS803	Computer Graphics and Multimedia	3	0	0	3	PEC	Specialization Core Course	Major (DSG)	Major (DSG)	Major (DSG)
4	CS804	AI/ML Technologies	3	0	0	3	PEC	Specialization Core Course	Major (DSG)	Major (DSG)	Major (DSG)
5	CS805	AI/ML Technologies Lab	0	0	4	2	PEC	Specialization Core Course	Major (DSG)	Major (DSG)	Major (DSG)

SPECIALIZATION CORE COURSES FOR B.Tech. CS: Specialization: Data Science										UCC Mapping	
S.No	Course Code	Course Name	L	T	P	Credits	UCC Mapping	UCC Mapping	UCC Mapping	UCC Mapping	UCC Mapping
1	CS901	Machine Learning	3	0	0	3	PEC	Specialization Core Course	Major (DSG)	Major (DSG)	Major (DSG)
2	CS902	Machine Learning Lab	0	0	4	2	PEC	Specialization Core Course	Major (DSG)	Major (DSG)	Major (DSG)
3	CS903	Data Mining and Analytics	3	0	0	3	PEC	Specialization Core Course	Major (DSG)	Major (DSG)	Major (DSG)
4	CS904	Data Mining and Analytics Lab	0	0	4	2	PEC	Specialization Core Course	Major (DSG)	Major (DSG)	Major (DSG)

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CSA 201	Big Data Analytics Lab	0	0	0	2	Specialization Core Course II Lab	Major (PCC)
SPECIALIZATION CORE COURSES FOR B.Tech CSE Specialization: Cyber Security							
CSA 202	Cryptography and Network Security	3	0	0	3	Specialization Core Course I	Major (PCC)
CSA 203	Application and Network Security Lab	0	0	0	2	Specialization Core Course I Lab	Major (PCC)
CSA 204	Introduction to Cyber Security	3	0	0	3	Specialization Core Course II	Major (PCC)
CSA 205	Cloud Security and Incident Response	3	0	0	3	Specialization Core Course III	Major (PCC)
CSA 206	Threat Hunting and Penetration Testing Lab	0	0	0	2	Specialization Core Course III Lab	Major (PCC)

PROGRAM ELECTIVES FOR B.Tech CSE									
S.No	Course Code	Course Name	L	T	P	Credits	Type		
1	CSA 301	Soft Computing	3	0	0	3	Program		
2	CSA 302	Introduction to Cyber Security	3	0	0	3	Program		
3	CSA 303	Game Design and Development	3	0	0	3	Program		
4	CSA 304	Information Retrieval Systems	3	0	0	3	Program		
5	CSA 305	Soft Computing Lab	0	0	0	2	Program		
6	CSA 306	Introduction to Cyber Security Lab	0	0	0	2	Program		
7	CSA 307	Game Design and Development Lab	0	0	0	2	Program		
8	CSA 308	Information Retrieval Systems Lab	0	0	0	2	Program		
9	CSA 309	Information Techniques	3	0	0	3	Program		
10	CSA 310	Data Mining and Warehousing	3	0	0	3	Program		
11	CSA 311	Artificial Intelligence	3	0	0	3	Program		
12	CSA 312	Parallel and Distributed Computing	3	0	0	3	Program		
13	CSA 313	Computer Vision	3	0	0	3	Program		
14	CSA 314	Artificial Language Processing	3	0	0	3	Program		
15	CSA 315	Wireless Communication Networks	3	0	0	3	Program		
16	CSA 316	Augmented and Virtual Reality	3	0	0	3	Program		
17	CSA 317	Blockchain Technology	3	0	0	3	Program		
18	CSA 318	Automation and Robotics	3	0	0	3	Program		
19	CSA 319	Quantum Computing	3	0	0	3	Program		
20	CSA 320	Digital Twin Systems	3	0	0	3	Program		

PROGRAM ELECTIVES FOR B.Tech CSE Specialization: Data Science									
S.No	Course Code	Course Name	L	T	P	Credits	Type		
1	CSA 401	Big Data Analytics and Visualization	3	0	0	3	Program		
2	CSA 402	Time Series Analysis and Forecasting	3	0	0	3	Program		
3	CSA 403	Machine Learning Data Analysis	3	0	0	3	Program		
4	CSA 404	Information Retrieval Systems	3	0	0	3	Program		
5	CSA 405	Big Data Analytics and Visualization Lab	0	0	0	2	Program		
6	CSA 406	Time Series Analysis and Forecasting Lab	0	0	0	2	Program		
7	CSA 407	Machine Learning Data Analysis Lab	0	0	0	2	Program		
8	CSA 408	Information Retrieval Systems Lab	0	0	0	2	Program		
9	CSA 409	Optimization Methods in Data Science	3	0	0	3	Program		
10	CSA 410	Statistical Modeling and Analysis	3	0	0	3	Program		
11	CSA 411	Deep Learning and Explainable AI	3	0	0	3	Program		
12	CSA 412	Bayesian Networks	3	0	0	3	Program		
13	CSA 413	Big Data Analytics and Visualization	3	0	0	3	Program		
14	CSA 414	Time Series Analysis and Forecasting	3	0	0	3	Program		
15	CSA 415	Machine Learning Data Analysis	3	0	0	3	Program		
16	CSA 416	Information Retrieval Systems	3	0	0	3	Program		
17	CSA 417	Big Data Analytics and Visualization Lab	0	0	0	2	Program		
18	CSA 418	Time Series Analysis and Forecasting Lab	0	0	0	2	Program		
19	CSA 419	Machine Learning Data Analysis Lab	0	0	0	2	Program		
20	CSA 420	Information Retrieval Systems Lab	0	0	0	2	Program		

PROGRAM ELECTIVES FOR B.Tech CSE Specialization: Cyber Security									
S.No	Course Code	Course Name	L	T	P	Credits	Type		
1	CSA 501	Secure Programming and Security	3	0	0	3	Program		
2	CSA 502	Reverse Engineering and Malware Analysis	3	0	0	3	Program		
3	CSA 503	Network Detection and Incident Response	3	0	0	3	Program		

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4	CSY315	Blockchain Security	3	0	0	0	3		
5	CSY316	Secure Coding, Penetration and White Hat Security Lab	0	0	0	0	0	7	
6	CSY317	Operating System Security Lab	0	0	0	0	0	7	
7	CSY318	Intrusion Detection and Prevention System Lab	0	0	0	0	0	7	
8	CSY319	Blockchain Security Lab	0	0	0	0	0	7	
9	CSY320	Mobile App Security and Reverse Engineering	3	0	0	0	0	3	
10	CSY321	Cloud Architecture and Security	3	0	0	0	0	3	
11	CSY322	Proprietary, Emerging Technology	3	0	0	0	0	3	
12	CSY323	Security Information and Event Management	3	0	0	0	0	3	
13	CSY324	System Control and Identity Management	3	0	0	0	0	3	
14	CSY325	Secure Software Engineering	3	0	0	0	0	3	
15	CSY326	Digital Forensics	3	0	0	0	0	3	
16	CSY327	AI for Cyber Security	3	0	0	0	0	3	
17	CSY328	Enterprise Security and Management	3	0	0	0	0	3	
18	CSY329	Mobile and Wireless Sensor Network Security	3	0	0	0	0	3	
19	CSY330	Threat Intelligence and Cyber Analytics	3	0	0	0	0	3	
20	CSY331	Cyber Law and Ethics	3	0	0	0	0	3	

The Integrated B.Tech.-M.Tech. (Computer Science and Engineering) programme shall follow the same course structure as the B.Tech. (Computer Science and Engineering) programme up to the Seventh Semester. After completion of the Seventh Semester, students shall opt for an M.Tech. specialization offered by the Department, subject to the applicable rules and availability. From the Eighth Semester onwards, students shall follow the curriculum of the selected M.Tech. (Computer Science and Engineering) specialization. Accordingly, the Eighth Semester shall be equivalent to the Second Semester of the respective M.Tech. specialization.

Exit at the end of first year, mandatory skill course module on Technical Communication and Computer Proficiency (Data Entry etc.), Civil / Mechanical Draftsmanship, Electrical maintenance / internship etc of 4 credit should be completed.

Exit after 2nd year of B.Tech. course must undergo skill modules on IT/Hardware Networking / MATLAB/ internship etc of 4 credit or Branch specific skill module.

At 3rd year and 4th year of B.Tech. is already Engineering specific, students who exit after 3-years may be awarded B.Sc(CS).

HSME/CAC/AEC/SEC and Open Elective courses shall be offered from a basket floated by the respective Departments prior to each semester.

UGC-COOP CATEGORY MAPPING - CONSOLIDATED CREDIT SUMMARY

Programme: B.Tech. in Computer Science and Engineering (CSE) - NEP 2020, 2020 and Specialization DS and Cyber Security, Department of CSE, SVKM's Deemed to be University

UGC-COOP Category	Description / Typical Usage	Credits (live)	% of Programme	UGC Suggested Target	Difference
Major (DS)	Discipline specific core courses - the primary field of study (maps to AICTE's PCC and named 'Specialization Core' courses)	110	65.5%	112	-2
Minor		0	0.0%	Added with Major	0
Multidisciplinary Course (MDC)	Foundational/allied science and open electives outside the core discipline (maps to AICTE's ESC, BSC, and OEC).	12	7.1%	9	3
Ability Enhancement Course (AEC)	Language, communication, and general HSMC electives.	9	5.4%	8	1
Skill Enhancement Course (SEC)	Hands-on, employability-focused skill courses (e.g. Career Skills).	15	8.9%	9	6
Value-Added Course (VAC)	Ethics, Indian Knowledge System, Constitution of India, and similar mandatory value courses.	6	3.6%	8	2
Research Project (Major)	Capstone Project 1 & 2 - culminating research/design work within the Major.	6	3.6%	12	6
Internship / Apprenticeship	Industrial Training and the final Internship.	10	6.0%	4	6
TOTAL		168	100.0%	162	6

Cross-check vs Programme Structure Total Credits (Sheet1 K131):

168

AICTE MAPPING WORK - SUMMARY

Credits are computed live from the AICTE Mapping column (Sheet1, column L) on the Programme Structure sheet. Indicate course's AICTE category there and this table reflects automatic.

AICTE Category	Category (Full Name)	Current Credits (live)	AICTE Suggested Target	Difference	Status
PCC	Professional Core Course	70	59	11	Above target x 11
PEC	Professional Elective Course	14	13	1	Above target x 1
ESC	Engineering Science Course	22	24	-2	Below target x 2
BSC	Basic Science Course	27	27	0	Below target x 0
HSMC	Humanities, Social Science & Mgmt. Course	15	16	-1	Below target x 1
OEC	Open Elective Course	9	9	0	Match target
Internship / Training / Apprenticeship	Capstone Project, Industrial Training & Internship	16	4	12	Above target
TOTAL		168	163	5	Above target by 5

Cross-check vs Programme Structure Total Credits (Sheet1 K231):

163

Approved by:  Date: 